

MLFF SIZE-FIDELITY2 Survivor-Requalification Specification

q=4..8 retrospective 3/10/30 fidelity authority

mdstats project

2026-08-16

Status

Gate: SIZE-FIDELITY2

Release: mdstats 0.20.207a0

Architecture revision: 74; **dependency-graph schema:** 56

Status: pre-migration control plane implemented; positive GPU qualification deferred to FINAL-GPU1

Next: TARGET-DATA2C-MVMIGRATE1

Calibration and continuation authority

For hard-qualified population `q_max`, calibrate every available `q` in `4..q_max`. Qualified sizes must be a contiguous suffix of (128, 256, 512, 1024, 2048, 4096, 8192, 16384). For each frozen optimizer seed, train every qualified size **once** on the common nominal 30-epoch schedule with halving disabled and evaluate epochs 3, 10, 30. Every `q` width is reconstructed from this same trajectory matrix, so work is `n_seed * q_max`, not separate `q`-specific campaigns.

Epoch 10 authenticates the epoch-3 checkpoint/optimizer/RNG parent; epoch 30 authenticates epoch 10. Foundation, evaluation role, TRAIN2 policy, training-run identity, and schedule identity cannot drift. Schedule progress is exactly 3/30, 10/30, 30/30, with strictly increasing update/exposure counts.

Survivor-recall and fixed-ceiling authority

For each seed/`q` width, the complete epoch-30 population defines the eventual two finalists using SIZE-HALVE2 deterministic practical-equivalence ordering. Reconstruct:

```
epoch 3:  q -> min(q,4)
epoch 10: 4 -> 2
epoch 30: complete q population is calibration truth
```

Both eventual finalists must survive epochs 3 and 10 for **every** seed/`q` width: required finalist recall is 1.0 at both boundaries. Winner recall is diagnostic only. If 16,384 remains materially superior to every smaller admissible candidate beyond the frozen practical-equivalence width, qualification fails as fixed-ceiling nonconvergence.

Monitor derivation and migration boundary

The monitor grid is 128, 256, 512, 1024. Every epoch-3 monitor score derives from the **same full-prediction digest** as the full-role score; later checkpoints cannot request monitor inference. Additional monitor-model inference count is therefore exactly zero. Recommend the smallest monitor whose epoch-3 promotion set exactly matches full-role promotion for every seed/`q` width and retains the eventual finalists.

`size_fidelity2_execution_plan` is content-addressed to SIZE-HALVE2 and included in the prepare restart receipt; blocked SIZE-HALVE2 yields a blocked no-work plan. Qualification reports require the exhaustive trajectory/checkpoint matrix, and unit fixtures cannot masquerade as production GPU evidence. Positive MACE/GPU execution remains deferred to FINAL-GPU1.

This gate changes no generated policy: revision-64 TARGET-DATA2C v4/TARGET-DATA2D v2 remain authoritative until TARGET-DATA2C-MVMIGRATE1 explicitly migrates policy after the required evidence exists.